Problem C
Two Greedy Customers

Time Limit: 3.0 seconds

A shop sells N types of products numbered from 1 to N. The price of product i is C_i.

Two customers, Alice and Bob, will visit the shop separately. When Alice purchases product i, she gains satisfaction U_i. When Bob purchases it, he gains satisfaction V_i. Initially, Alice has A units of money, and Bob has B units of money.

Before they visit, you, a shop clerk, choose an ordering of the N product types. Each customer independently considers the product types one by one in the chosen order and makes purchases according to the following rules.

- The customer inspects the products one by one from left to right.
- If the price of the current product is not greater than the amount of money the customer has at that moment, the customer always purchases one item of that product and pays its price.
- Otherwise, the customer does not purchase that product.

Each product is available in sufficient quantity, and the purchases made by Alice and Bob do not affect each other.

The score of each customer is the sum of the satisfaction gained from the products purchased by that customer. Determine the maximum possible sum of the scores of Alice and Bob by arranging the products appropriately.

Input

The input consists of a single test case in the following format.

$$
\begin{aligned}
&N\ A\ B\\
&C_1\ U_1\ V_1\\
&C_2\ U_2\ V_2\\
&\vdots\\
&C_N\ U_N\ V_N
\end{aligned}
$$

The integer N is the number of product types ($1 \leq N \leq 200$). The integers A and B are the initial amounts of money held by Alice and Bob, respectively ($1 \leq A, B \leq 120$).

For each $i = 1, \ldots, N$, the integer C_i is the price of product i ($1 \leq C_i \leq 120$). The integers U_i and V_i are the satisfaction gained by Alice and Bob, respectively, when they purchase product i ($1 \leq U_i, V_i \leq 10^9$).

Output

Output the maximum possible sum of the scores of Alice and Bob in one line.

Sample Input 1	Sample Output 1
2 6 8 3 10 1 5 8 10	21

Sample Input 2	**Sample Output 2**
2 1 1 1 5 1 1 1 10	11

Sample Input 3	**Sample Output 3**
3 5 6 4 1 100 3 100 1 2 10 10	220

If the products are arranged in the order $1, 2$ in the first sample, Alice purchases only product 1 and gets a score of 10, while Bob purchases both products and gets a score of 11. The total score is 21, which is the maximum possible.